

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-IV EXAMINATION – SUMMER 2026

Subject Code: 3140702

Date: 19-05-2026

Subject Name: Operating System

Time: 10:30 AM TO 01:00 PM

Total Marks: 70

Instructions:

1. Attempt all questions.
 2. Make suitable assumptions wherever necessary.
 3. Figures to the right indicate full marks.
 4. Simple and non-programmable scientific calculators are allowed.
-

	Marks
Q.1 (a) List any four functions of operating system?	03
(b) Explain the essential properties of i) Batch system ii) Time sharing iii) Real time iv) Distribute	04
(c) What is Semaphore? Give the implementation of Bounded Buffer Producer Consumer Problem using Semaphore.	07
Q.2 (a) Discuss the major goals of I/O software	03
(b) What is scheduling? Explain the types of schedulers.	04
(c) List out various criteria for good process scheduling algorithms. Illustrate non-preemptive priority scheduling algorithm.	07

OR

(c) What is monitor? Explain solution for producer-consumer problem using monitor.	07
Q.3 (a) What is Process? Give the difference between a process and a program.	03
(b) What is Process State? Explain different states of a process with various queues generated at each stage.	04
(c) Briefly explain and compare, fixed and dynamic memory partitioning schemes.	07

OR

(a) Define the difference between preemptive and nonpreemptive scheduling.	03
(b) Differentiate user level and kernel level thread.	04
(c) Write a bounded-buffer monitor in which the buffers (portions) are embedded within the monitor itself.	07

- Q.4** (a) What is thread? Explain classical thread model. **03**
(b) What is external fragmentation? Explain the solution to external fragmentation. **04**
(c) Illustrate Readers and Writers IPC problem with solution. **07**

OR

- Q.4** (a) Discuss the major goals of I/O software **03**
(b) Explain the terms related to IPC: **04**
i) Race condition ii) Critical section iii) Mutual exclusion iv) Semaphores
(c) Write short note on RAID levels. **07**

- Q.5** (a) Write a short note: Unix kernel. **03**
(b) What are components of Linux systems? **04**
(c) Explain process control block with all parameters. **07**

OR

- Q.5** (a) What is deadlock? List the conditions that lead to deadlock. **03**
(b) Compare virtual machine and non virtual machine. **04**
(c) Explain Disk arm scheduling algorithm. **07**
